



Lullaby Baby Warmer



Low-cost smartphone



Low-cost tablets



Tata Nano

machine and the Lullaby Baby Warmers have won international acclaim. Now these products are exported to several countries including developed nations," says Srinivas.

Low-cost tablets. Low-cost tablets, such as Aakash with basic features and open source software, are also becoming popular these days. Rather than using these tablets as general-purpose machines that have all the functions, customised versions, which have only the features important to a group, can be used. This is another method of making a product cost-effective.

Mobile-controlled irrigation. Nano Ganesh is an application developed by Pune-based Ossian Agro Automation. It has been in use for around five years now, and continues to be popular in Maharashtra. It has even been recognised by the World Bank as a notable technology for developing nations, due to its simplicity and affordability. Nano Ganesh, which also won a Nokia innovation award, allows farmers to turn their irrigation pumps on and off by remotely using a mobile phone. A mobile modem is placed on the pump and connected to a basic mobile phone. This allows remote control of the pump's starter panel. When the

farmer calls this phone from any location, a tone is heard if the motor is on, and there is no sound if the motor is off. The farmer can then turn the motor on or off through a 440-volt starter switch that responds to a code number sent via SMS from the farmer's mobile phone to the phone attached to the motor. Nano Ganesh can be attached to any electrical starter and motor pump. It costs Rs 2800 plus taxes, and setting it up is easy and automatic. Users do not have to be literate to understand the simple tones. Overall, it makes sense for the agro community.

Low-cost smartphones. The initial smartphones arrived at a price-point of ₹ 30,000, which was way too much for an average Indian. So, how best can this product be optimised for a low-income group? "The high-end phones came with support for multi-band 3G roaming capability. For an average Indian, this is not needed. So, costs are saved by removing this feature. Optimising the software for lower memory footprint gives the opportunity to save some more costs. Earlier, systems used NAND Flash boot while still having an SD card. Products for the low-end markets pushed technology to use an SD card boot, thus saving the cost of

additional memory. While a natural drop in the prices helped this market, lower-resolution displays and using a single speaker to give virtualisation effects all helped address the needs of this market. Adding dual-SIM capability added the icing on the cake for this segment," explains Srinivas beautifully!

Tata Nano. This is a moderately successful car, with decent value and low cost. "The car is designed for those who can afford a two-wheeler but can stretch a bit to get the comfort of a car," says Srinivas. "This meant the car had to have all the necessary features but be priced effectively. This implied optimising the space inside the car, which also meant re-positioning the engine accordingly, for both efficiency and space. This needed a complete re-design of the engine and chassis. Tata Motors also went ahead and re-designed the doors and windows for lower weight so as to improve the efficiency. Components have been manufactured indigenously to the extent possible to save on the import costs."

Cost-effective computing. In 2011, the government of Rajasthan focussed its attention on a massive computer education program, which involved setting up computing labs in around 2000 schools spread across the state's 33 districts. If implemented the traditional way, it would have meant an unbearable initial investment atop the ongoing costs in support, maintenance, power supply and so on. Exploring alternatives to this, they came across NComputing's practical and cost-effective computing solution. NComputing offers desktop virtualisation solutions, which give access to computing at a fraction of the cost of traditional server-PC based networks. This enables educational institutions, hospitals and rural BPOs to realise their dreams at lower price points. Typically, an average computer user uses merely 5-10 per cent of the PC's capacity; the rest (approx. 90 per cent) goes waste. In the case of the Rajasthan schools too, their choice of NComputing solution was based